

Fluo-suite nuclei-selection survey: connected-nuclei counts vs. dipolar cutoff

1 Method

For each molecule in the fluo_suite dataset (`circ_sim/data/fluo_suite/ORCA_NMR_summary_merged.xlsx`), a chemistry-constrained candidate pool is built from the full set of $^1\text{H}/^{13}\text{C}/^{19}\text{F}$ nuclei as follows (see `circ_sim/scripts/zulf_numerics/fluo_suite/moment_convergence.py`):

1. **Seed:** all ^{19}F atoms, plus a single representative carbon, reflecting natural ^{13}C abundance ($\sim 1.1\%$): at most one carbon is isotopically active in a given molecule at a time.
2. **Candidate pool:** all non-exchangeable protons, identified as H atoms with at least one 1-bond C–H coupling in the dataset (a data-driven proxy for “bonded to carbon, not to O/N”). Exchangeable protons ($-\text{OH}$, $-\text{NH}_2$, ...) are excluded.

Multiple fluorinated anchors. A “fluorinated anchor” is a carbon directly (1-bond) coupled to at least one F atom. Several molecules in the suite contain more than one *chemically separate* anchor (e.g. two independent CF_2 or CF_3 groups on different rings). Since natural abundance only ever makes one carbon isotopically active at a time, each anchor is an equally valid, mutually exclusive scenario for “which carbon is the ^{13}C this time” – there is no principled reason to keep only the most strongly J -coupled one. When a molecule has $N \geq 1$ such anchors, N separate versions of it are therefore carried through the rest of the analysis, one per anchor, labelled with a letter suffix (a, b, c, ...) ordered by decreasing cumulative $J(\text{C}, \text{F})$ – version “a” reproduces the single-anchor choice used in the original (pre-versioning) survey. In every version the seed still contains *all* ^{19}F atoms in the molecule, not just the ones bonded to that version’s anchor: an F-detected experiment observes every ^{19}F resonance simultaneously regardless of which carbon happens to be ^{13}C , so only the local splitting/connectivity pattern near the labelled anchor differs between versions, not the detection channel itself.

The *maximal number of chemically/NMR-relevant nuclei* for a molecule is the size of this constrained set (seed + candidate pool), independent of any distance cutoff – the ceiling on how many nuclei could ever be retained under these two chemistry constraints alone.

Within that constrained set, reachability from the seed is then determined by a combined coherent+dipolar coupling graph:

- *Coherent edges:* 1-bond/2-bond J -couplings from the ORCA/PySCF dataset, weight $2\pi|J_{ij}|$ (rad/s).
- *Dipolar edges:* through-space coupling computed from atomic coordinates for *every* pair within the stated cutoff, weight $b_{ij}^2 \cdot (2\tau_c/5)$ (rad/s), with $b_{ij} = -(\mu_0/4\pi)\gamma_i\gamma_j\hbar/r_{ij}^3$ and $\tau_c = 10^{-10}\text{ s}$ – the same spectral-density prefactor used in the rank-2 Lindblad jump operators elsewhere in this codebase, so the two channels are combined in matched units rather than an arbitrary rescaling.

A candidate atom is counted as *connected* if there exists a path of nonzero-weight edges (either channel) back to the seed. This reachability criterion is a graph-only upper bound: it does not verify that a reachable atom measurably perturbs the ^{19}F -detected signal (see the moment-convergence discussion in `moment_convergence.py` for a tighter, dynamics-based criterion).

2 Results

Table 1 reports, for every molecule in the suite (mol4, Pibrentasvir, was not computed at the ORCA/PySCF stage and is omitted) and every fluorinated-anchor version thereof, the anchor carbon (and the F atom(s) it is 1-bond coupled to), the total number of NMR-active nuclei, the maximal chemistry-relevant set size, and the number of nuclei connected to the seed under the combined graph at cutoffs of 5.0, 3.0, 2.5, and 2.0 Å. 12 of the 21 molecules have more than one fluorinated anchor (up to 4, for Atogepant), giving 36 rows in total.

ID	Drug name	Anchor	Total nuclei (H+C+F)	Max relevant (ceiling)	Connected nuclei			
					5.0 Å	3.0 Å	2.5 Å	2.0 Å
mol1	Lorlatinib	C6 (F8)	41	19	19	15	13	4
mol2a	Voxilaprevir	C40 (F54,F55)	96	54	54	48	21	8
mol2b		C13 (F14,F15)	96	54	54	48	21	7
mol3a	Glecaprevir	C18 (F19,F20)	88	48	48	44	26	7
mol3b		C37 (F38,F39)	88	48	48	44	30	7
mol5a	Belzutifan	C16 (F22)	32	15	15	9	9	8
mol5b		C8 (F23)	32	15	15	9	7	6
mol5c		C7 (F24)	32	15	15	9	7	7
mol6	Pexidartinib	C25 (F26,F27,F28)	38	17	17	9	5	4
mol7	Tecovirimat	C23 (F24,F25,F26)	37	18	8	8	5	4
mol8a	Ivosidenib	C29 (F31,F32)	53	25	25	23	8	8
mol8b		C11 (F15)	53	25	25	23	6	6
mol9a	Tezacaftor	C33 (F35,F36)	56	27	27	19	4	4
mol9b		C10 (F19)	56	27	27	19	19	5
mol10	Emtricitabine	C9 (F15)	19	9	9	9	9	3
mol11	Gemcitabine	C9 (F16,F17)	22	10	10	10	7	5
mol12	Sofosbuvir	C15 (F27)	52	28	28	21	9	7
mol13	Tedizolid phosphate	C13 (F30)	34	16	13	9	9	3
mol14a	Cabotegravir	C23 (F26)	38	18	18	8	5	5
mol14b		C21 (F27)	38	18	18	8	4	4
mol15	Linezolid	C13 (F23)	37	21	21	18	18	3
mol16	Eravacycline	C9 (F39)	59	26	26	16	9	3
mol17a	Atogepant	C9 (F10,F11,F12)	58	28	28	19	17	9
mol17b		C38 (F41)	58	28	28	19	18	8
mol17c		C39 (F40)	58	28	28	19	17	7
mol17d		C35 (F42)	58	28	28	19	18	8
mol18a	Letermovir	C8 (F9,F10,F11)	61	32	32	32	10	5
mol18b		C16 (F20)	61	32	32	32	11	6
mol19	Siponimod	C24 (F25,F26,F27)	67	38	38	20	20	4
mol20a	Fluticasone furoate	C22 (F23)	59	32	32	32	28	7
mol20b		C14 (F34)	59	32	32	32	28	9
mol20c		C6 (F36)	59	32	32	32	28	9
mol21a	Fluocinolone acetonide	C4 (F30)	56	31	31	31	31	6
mol21b		C20 (F29)	56	31	31	31	31	6
mol22a	Dutasteride	C19 (F20,F21,F22)	63	35	35	35	35	7

ID	Drug name	Anchor	Total nuclei (H+C+F)	Max relevant (ceiling)	Connected nuclei			
					5.0 Å	3.0 Å	2.5 Å	2.0 Å
mol22b		C23 (F24,F25,F26)	63	35	35	35	35	7

Table 1: Connected-nuclei counts under the combined coherent+dipolar graph, by cutoff, for every fluorinated-anchor version of every molecule in the suite. “Anchor” gives the version’s ^{13}C atom and the F atom(s) it is 1-bond coupled to. “Max relevant” is the chemistry-constrained ceiling (seed + non-exchangeable-H candidate pool) and, like “Total nuclei”, is identical across versions of the same molecule (it depends only on the number of F atoms and non-exchangeable H’s, not on which carbon is chosen). At 5.0 Å the graph reaches the ceiling for most molecule/anchor versions; mol7 (Tecovirimat) and mol13 (Tedizolid phosphate) are exceptions where part of the candidate pool sits 10+ Å from the seed even at this cutoff (an elongated molecule with the anchor at one end and remaining protons on a distant ring), not a data or algorithm artifact. By 2.5 Å the different anchors of the same molecule already start to diverge noticeably (e.g. mol9a: 4 vs. mol9b: 19; mol3a: 26 vs. mol3b: 30), well before either reaches its 2.0 Å value – so 2.5 Å is where anchor-dependence first becomes significant, not just a smaller version of the 5.0 Å picture. At 2.0 Å almost all versions collapse to a small (3–9-nuclei) immediate neighborhood of the anchor, where different anchors of the same molecule continue to disagree (e.g. mol5: 8/6/7, mol17: 9/8/7/8, mol20: 7/9/9) – consistent with tight cutoffs only reaching each anchor’s own local environment.

3 Filtered cutoff selection

Table 2 applies a simple rule to pick, per molecule/anchor version, a small pair of cutoffs to actually use rather than the full 5.0/3.0/2.5/2.0 Å sweep: always keep 5.0 Å, then take the *next* cutoff (scanning 3.0 Å, then 2.5 Å, then 2.0 Å) at which the connected-nuclei count first changes from its 5.0 Å value. If that changed value is ≥ 9 it is kept as the second cutoff; if it is < 9 , no second cutoff is kept for that row (only 5.0 Å is reported) – scanning does not continue past the first change point even if a later, untested-in-this-rule cutoff might happen to be ≥ 9 , since the count is monotonically non-increasing as the cutoff tightens, so nothing later in the scan could exceed a value already below the floor. Rows whose 5.0 Å count itself is < 9 are dropped entirely, since no cutoff in the sweep could satisfy the floor – this removes mol7 (Tecovirimat: 8 at 5.0 Å already), leaving 35 of the 36 rows.

ID	Drug name	Anchor	Total nuclei (H+C+F)	Connected @5.0 Å	Second cutoff (Å)	Connected @second cutoff
mol1	Lorlatinib	C6 (F8)	41	19	3.0	15
mol2a	Voxilaprevir	C40 (F54,F55)	96	54	3.0	48
mol2b		C13 (F14,F15)	96	54	3.0	48
mol3a	Glecaprevir	C18 (F19,F20)	88	48	3.0	44
mol3b		C37 (F38,F39)	88	48	3.0	44
mol5a	Belzutifan	C16 (F22)	32	15	3.0	9
mol5b		C8 (F23)	32	15	3.0	9
mol5c		C7 (F24)	32	15	3.0	9
mol6	Pexidartinib	C25 (F26,F27,F28)	38	17	3.0	9
mol8a	Ivosidenib	C29 (F31,F32)	53	25	3.0	23

ID	Drug name	Anchor	Total nuclei (H+C+F)	Connected @5.0 Å	Second cutoff (Å)	Connected @second cutoff
mol8b		C11 (F15)	53	25	3.0	23
mol9a	Tezacaftor	C33 (F35,F36)	56	27	3.0	19
mol9b		C10 (F19)	56	27	3.0	19
mol10	Emtricitabine	C9 (F15)	19	9	–	–
mol11	Gemcitabine	C9 (F16,F17)	22	10	–	–
mol12	Sofosbuvir	C15 (F27)	52	28	3.0	21
mol13	Tedizolid phosphate	C13 (F30)	34	13	3.0	9
mol14a	Cabotegravir	C23 (F26)	38	18	–	–
mol14b		C21 (F27)	38	18	–	–
mol15	Linezolid	C13 (F23)	37	21	3.0	18
mol16	Eravacycline	C9 (F39)	59	26	3.0	16
mol17a	Atogepant	C9 (F10,F11,F12)	58	28	3.0	19
mol17b		C38 (F41)	58	28	3.0	19
mol17c		C39 (F40)	58	28	3.0	19
mol17d		C35 (F42)	58	28	3.0	19
mol18a	Letermovir	C8 (F9,F10,F11)	61	32	2.5	10
mol18b		C16 (F20)	61	32	2.5	11
mol19	Siponimod	C24 (F25,F26,F27)	67	38	3.0	20
mol20a	Fluticasone furoate	C22 (F23)	59	32	2.5	28
mol20b		C14 (F34)	59	32	2.5	28
mol20c		C6 (F36)	59	32	2.5	28
mol21a	Fluocinolone acetonide	C4 (F30)	56	31	–	–
mol21b		C20 (F29)	56	31	–	–
mol22a	Dutasteride	C19 (F20,F21,F22)	63	35	–	–
mol22b		C23 (F24,F25,F26)	63	35	–	–

Table 2: Filtered cutoff selection: 5.0 Å plus the next cutoff at which the connected-nuclei count changes, subject to a floor of 9 nuclei (see text). “–” means no second cutoff qualifies: either the count never changes down to 2.0 Å (mol10, mol11, mol14a/b), or the first change already drops to or below the floor (mol21a/b at 2.0 Å goes to 6; mol22a/b to 7). For most rows the qualifying second cutoff is 3.0 Å; mol18a/b and mol20a/b/c are the only rows where the count survives unchanged through 3.0 Å and only breaks at 2.5 Å.